

```

if (main_time % 4) == 2) {
    top -> a = 0;
    top -> b = 1;
}

if (main_time % 4) == 3) {
    top -> a = 1;
    top -> b = 1;
}

top -> eval();

if (tfp) tfp -> dump(main_time);

main_time++;
}

top -> final();

if (tfp) tfp -> close();

delete top;

exit(0);
}

```

The while loop handles the different cases for various combinations of the input. Steps to compile, build and test the half-adder example can be automated in a Makefile.

TARGET=halfadder

```

all:
    verilator -cc $(TARGET).v --exe sim_main.cpp --trace

build:
    make -j -C obj_dir -f V$(TARGET).mk V$(TARGET)

test:
    ./obj_dir/V$(TARGET)

clean:
    rm -rf obj_dir /*.vcd

```

Sources can be compiled with Verilator using make.

\$ make

verilator -cc halfadder.v --exe_main.cpp --trace

The generated C++ code is then built using the following:

\$ make build

```

make -j -C obj_dir -f Vhalfadder.mk Vhalfadder
make[1]: Entering directory '/home/guest/halfadder/obj_dir'
g++ -I. -MMD -I/usr/share/verilator/include \
-I/usr/share/verilator/include/vltstd -DVL_PRINTF=printf \
-DVM_TRACE=1 -DVM_COVERAGE=0 \
-c -o main.o ../main.cpp

```



Fig. 1: Waveform

```

g++ -I. -MMD -I/usr/share/verilator/include \
-I/usr/share/verilator/include/vltstd -DVL_PRINTF=printf \
-DVM_TRACE=1 -DVM_COVERAGE=0 \
-c -o verilated.o /usr/share/verilator/include/
verilated.cpp

g++ -I. -MMD -I/usr/share/verilator/include \
-I/usr/share/verilator/include/vltstd -DVL_PRINTF=printf \
-DVM_TRACE=1 -DVM_COVERAGE=0 -c -o verilated_vcd.o \
/usr/share/verilator/include/verilated_vcd.o \
/usr/bin/perl /usr/share/verilator/bin/verilator_incluser \
Vhalfadder.cpp > Vhalfadder_ALLc1s.cpp
/usr/bin/perl /usr/share/verilator/bin/verilator_incluser \
Vhalfadder_Trace.cpp Vhalfadder_Syms.cpp \
Vhalfadder_Trace_Slow.cpp > Vhalfadder_ALLsup.cpp
g++ -I. -MMD -I/usr/share/verilator/include \
-I/usr/share/verilator/include/vltstd -DVL_PRINTF=printf \
-DVM_TRACE=1 -DVM_COVERAGE=0 \
-c -o Vhalfadder_ALLc1s.o Vhalfadder_ALLc1s.cpp
g++ -I. -MMD -I/usr/share/verilator/include \
-I/usr/share/verilator/include/vltstd -DVL_PRINTF=printf \
-DVM_TRACE=1 -DVM_COVERAGE=0 \
-c -o Vhalfadder_ALLsup.o Vhalfadder_ALLsup.cpp
Archiving Vhalfadder_ALL.a ...
ar r Vhalfadder_ALL.a Vhalfadder_ALLc1s.o Vhalfadder_
ALLsup.o
ar: creating Vhalfadder_ALL.a
ranlib Vhalfadder_ALL.a
g++ main.o verilated.o verilated_vcd.o Vhalfadder_ALL.a \
-o Vhalfadder -lm -lstdc++ -DVL_PRINTF=printf \
make[1]: Leaving directory '/home/guest/halfadder/obj_dir'

```

You can test the code with the following:

\$ make test

./obj_dir/Vhalfadder

This produces counter.vcd file, which can be viewed in GTK-Wave. A screenshot of the waveform is shown in Fig. 1.

Install GTKwave before viewing the waveform.

You may also refer to Verilator manual at www.veripool.org/projects/verilator/wiki/Manual-verilator for more options and examples. ●



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EFY Note

The source codes of this project are included in this month's EFY DVD and are also available for free download at source.efymag.com